Test Booklet Code

Test Booklet No. ←

ENGLISH

02

NEET RANKERS TEST SERIES-2025

Do not open this Test Booklet until you are asked to do so.

SYLLABUS:

Physics:	Mechanical Properties of Solids, Mechanical Properties of Fluids, Thermal Properties of Matter, Thermodynamics, Kinetic Theory, Oscillations, Waves
Chemistry:	Redox Reaction, Organic Chemistry- Some Basic Principles and Techniques, Purification and characterization of organic compounds, Hydrocarbon
Botany:	Cell: The unit of life, Cell cycle and cell division, Photosynthesis in higher plants, Respiration in plants, Plant growth and development
Zoology:	Breathing and Exchange of Gases, Body Fluids and Circulation, Excretory Products and their Elimination, Locomotion and Movements, Neural control and Coordination, Chemical Control and Integration

Important Instructions:

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
2. The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Botany and Zoology)**. 50 questions in each subject are divided into **two Sections (A and B)** as per details given below:
   (a) **Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 and 185). All questions are compulsory.
   (b) **Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150, and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject.
   Candidates are advised to read all 15 questions in each subject of Section B before they start attempting the question paper. In the event of a candidate attempting more than ten questions, **the first ten questions answered by the candidate shall be evaluated.**
3. Each question carries 4 marks. For each correct response, the candidate will get **4 marks**. For each incorrect response, **one** mark will be deducted from the total scores. The **maximum marks are 720.**
4. **Use Blue/Black Ball Point Pen only** for writing particulars on this page/marking responses on Answer Sheet.
5. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
6. On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
7. **The CODE for this Booklet is 02. Make sure that the CODE printed on the Original Copy of the Answer Sheet is the same as that on this Test Booklet.** In case of discrepancy, the candidate should immediately report the matter to the Invigilator for replacement of both the Test Booklet and the Answer Sheet.
8. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
9. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
10. Each candidate must show on-demand his/her Admit Card to the Invigilator
11. No candidate, without special permission of the centre Superintendent or Invigilator, would leave his/her seat.
12. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet twice. Cases, **where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
13. Use of Electronic/Manual Calculator is prohibited.
14. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
15. **No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
16. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
17. Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of Scribe or not.

Name of the Candidate (in Capitals): ______________________________

Roll Number : In figures ______________________________

: In words ______________________________

Centre of Examination (in Capitals): ______________________________

Candidate's Signature: ____________________ **Invigilator's Signature:** ____________________

Facsimile Signature stamp of Centre Superintendent

Physics : Section-A (Q. No. 1 to 35)

1. Under the same load, wire A having length 5.0 m and cross–section area 2.5×10^{-5} m^2 stretches uniformly by the same amount as another wire B of length 6.0 m and a cross–section of 3.0×10^{-5} m^2 stretches.
The ratio of the Young's modulus of wire A to that of wire B will be
(1) 1 : 2 (2) 1 : 4
(3) 1 : 1 (4) 1 : 10

2. The rms speed of oxygen molecule in a vessel at particular temperature is $\left(1+\frac{5}{x}\right)^{1/2} v$, where v is the average speed of the molecule. The value of x will be $\left(\text{Take, } \pi=\frac{22}{7}\right)$
(1) 28 (2) 8
(3) 4 (4) 27

3. 10 gm of ice cubes at $0°C$ are released in a tumbler (water equivalent 55 g) at $40°C$. Assuming that negligible heat is taken from the surroundings the temperature of water in the tumbler becomes nearly (L = 80 cal/g):
(1) 31°C (2) 22°C
(3) 19°C (4) 15°C

4. The length of a metal wire is l_1, when the tension in it is T_1 and is l_2 when the tension is T_2. The natural length of the wire is
(1) $\sqrt{l_1 l_2}$ (2) $\frac{l_1T_2 - l_2T_1}{T_2 - T_1}$
(3) $\frac{l_1T_2 + l_2T_1}{T_2 + T_1}$ (4) $\frac{l_1 + l_2}{2}$

5. Two wires are made of the same material and have the same volume. The first wire has cross-sectional area A and the second wire has cross-sectional area 3 A. If the length of the first wire is increased by l on applying a force F, how much force is needed to stretch the second wire by the same amount?
(1) 4 F (2) 6 F
(3) 9 F (4) F

6. If S is stress and Y is Young's modulus of material of a wire, the energy stored in the wire per unit volume is
(1) $2S^2Y$ (2) $\frac{S^2}{2Y}$
(3) $\frac{2Y}{S^2}$ (4) $\frac{S}{2Y}$

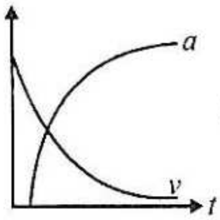
7. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A: When you squeeze One end of a tube to get toothpaste out from the other end, Pascal's principle is observed.
Reason R: A change in the pressure applied to enclosed incompressible fluid is transmitted undiminished to every portion of the fluid and to the walls of its container.
In the light of the above statements, choose the **correct** answer from the options given below.
(1) **A** is true but **R** is false.
(2) **A** is false but **R** is true.
(3) Both **A** and **R** are true and **R** is the correct explanation of **A**.
(4) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

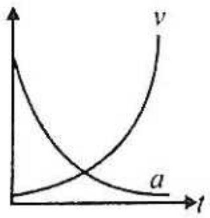
8. When a block of mass M is suspended by a long wire of length L, the length of the wire becomes $(L + l)$. The elastic potential energy stored in the extended wire is:
(1) Mgl (2) MgL
(3) $\frac{1}{2}$Mgl (4) $\frac{1}{2}$MgL

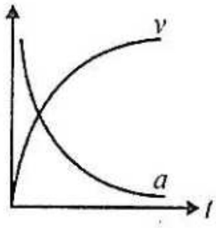
9. A string is stretched between fixed points separated by 75.0 cm. It is observed to have resonant frequencies of 420 Hz and 315 Hz. There are no other resonant frequencies between these two. Then, the lowest resonant frequency for this string is

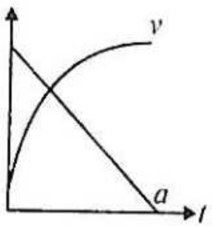
(1) 105 Hz (2) 1.05 Hz
(3) 1050 Hz (4) 10.5 Hz

10. Which of the following options correctly describes the variation of the speed V and acceleration a of a point mass falling vertically in a viscous medium that applies a force $F = -kV$, where k is a constant, on the body? (Graphs are schematic and not drawn to scale)

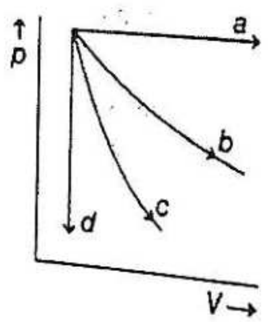
(1) 

(2) 

(3) 

(4) 

11. The given diagram shows four processes, i.e. isochoric, isobaric, isothermal and adiabatic. The correct assignment of the processes, in the same order is given by

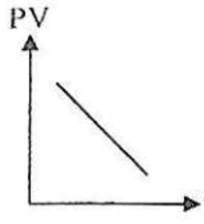


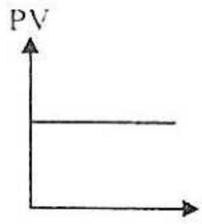
(1) *dabc* (2) *adbc*
(3) *dacb* (4) *adcb*

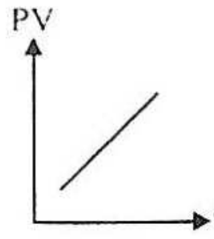
12. Each side of a box made of metal sheet in cubic shape is a at room temperature T; the coefficient of linear expansion of the metal sheet is α. The metal sheet is heated uniformly, by a small temperature ΔT, so that its new temperature is $T + \Delta T$. Calculate the increase in the volume of the metal box.

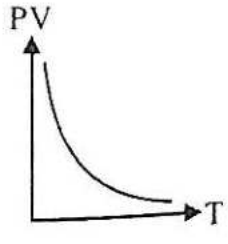
(1) $4\pi a^3 \alpha\Delta T$ (2) $4a^3 \alpha\Delta T$
(3) $\frac{4}{3}\pi a^3\alpha\Delta T$ (4) $3a^3 \alpha\Delta T$

13. Which of the following graphs represent the behaviour of an ideal gas? (Symbols have their usual meanings.)

(1) 

(2) 

(3) 

(4) 

14. The height of victoria falls is 63 m. What is the difference in temperature of water at the top and at the bottom of fall?
[Given, 1 cal = 4.2 J and specific heat of water = 1 cal g^{-1} °C^{-1}]

(1) 0.147°C (2) 14.76°C
(3) 1.476°C (4) 0.014°C

15. If a simple harmonic motion is represented by its $\frac{d^2x}{dt^2}+\alpha x=0$, its time period is

(1) $\frac{2\pi}{\alpha}$ (2) $\frac{2\pi}{\sqrt{\alpha}}$

(3) $2\pi\alpha$ (4) $2\pi\sqrt{\alpha}$

16. A liquid does not wet the solid surface if angle of contact is:

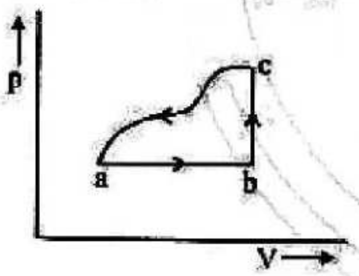
(1) Equal to 60°

(2) Greater than 90°

(3) Zero

(4) Equal to 45°

17. A sample of an ideal gas is taken through the cyclic process *abca* as shown in the figure. The change in the internal energy of the gas along the path *ca* is –180 J. The gas absorbs 250 J of heat along the path *ab* and 60 J along the path *bc*. The work done by the gas along the path *abc* is



(1) 120 J (2) 130 J

(3) 100 J (4) 140 J

18. One mole of rigid diatomic gas performs a work of $Q/5$ when heat Q is supplied to it. The molar heat capacity of the gas during this transformation is $\frac{xR}{8}$. The value of x is:

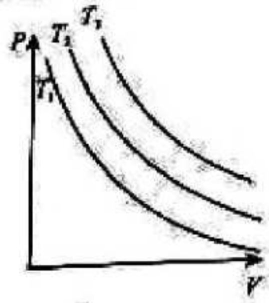
[R = universal gas constant]

(1) 25 (2) 50

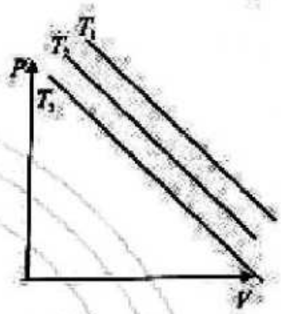
(3) 75 (4) 100

19. In an isothermal change, the change in pressure and volume of a gas can be represented for three different temperature $T_3 > T_2 > T_1$ as

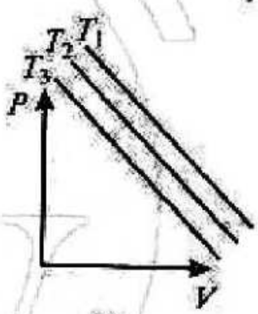
(1)



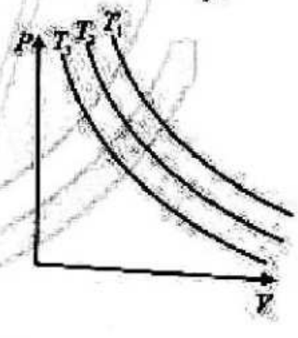
(2)



(3)



(4)



20. If 1000 droplets of water of surface tension 0.07 N/m, having same radius 1 mm each, combine to form a single drop. In the process, the released surface energy is $\left(\text{Take}, \pi = \frac{22}{7}\right)$

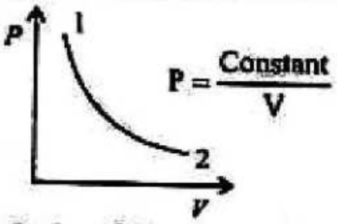
(1) 7.92×10^{-6} J

(2) 8.8×10^{-5} J

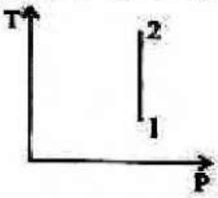
(3) 9.68×10^{-4} J

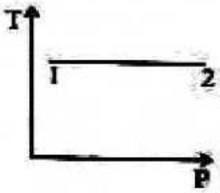
(4) 7.92×10^{-4} J

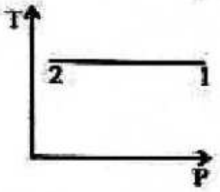
21. For the $p-V$ diagram given for an ideal gas,

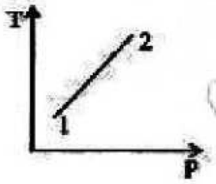


out of the following which one correctly represents the T – P diagram?

(1) 

(2) 

(3) 

(4) 

22. The temperature of an ideal gas is increased from 200 K to 800 K. If r.m.s. speed of gas at 200 K is v_0 Then, r.m.s. speed of the gas at 800 K will be

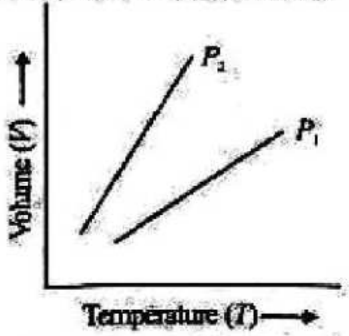
(1) $v_0/4$ (2) v_0
(3) $4v_0$ (4) $2v_0$

23. The initial pressure and volume of an ideal gas are p_0 and V_0. The final pressure of the gas when the gas is suddenly compressed to volume $\frac{V_0}{4}$ will be

(Given, γ = ratio of specific heats at constant pressure and at constant volume)

(1) $4p_0$ (2) p_0
(3) $p_0(4)^{\frac{1}{\gamma}}$ (4) $p_0(4)^{\gamma}$

24. For a perfect gas, two pressures P_1 and P_2 are shown in figure, The graph shows



(1) $P_1 > P_2$ (2) $P_1 < P_2$
(3) $P_1 = P_2$ (4) $P_1 = 2P_2$

25. An ideal gas is expanding such that pT^3 = constant. The coefficient of volume expansion of the gas is

(1) $1/T$ (2) $2/T$
(3) $4/T$ (4) $3/T$

26. A rod of length L at room temperature and uniform area of cross-section A, is made of a metal having coefficient of linear expansion $\alpha/°C$. It is observed that an external compressive force F, is applied on each of its ends, prevents any change in the length of the rod, when its temperature rises by ΔTK. Young's modulus Y, for this metal is

(1) $\frac{2F}{A\alpha\Delta T}$ (2) $\frac{F}{A\alpha\Delta T}$
(3) $\frac{F}{2A\alpha\Delta T}$ (4) $\frac{F}{A\alpha(\Delta T - 273)}$

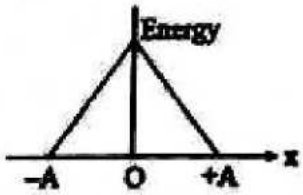
27. If the length of a wire is made double and radius is halved of its respective values, then the Young's modulus of the material of the wire will

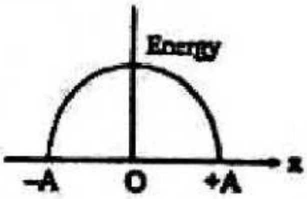
(1) remain same
(2) become 8 times its initial value
(3) become 1/4 of its initial value
(4) become 4 times of its initial value

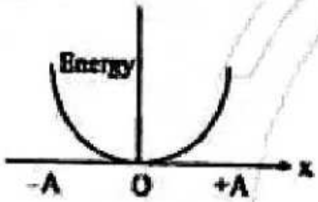
28. A diatomic gas having $C_p = 7/2\,R$ and $C_v = 5/2R$, is P heated at constant pressure. The ratio $dU : dQ : dW$ is

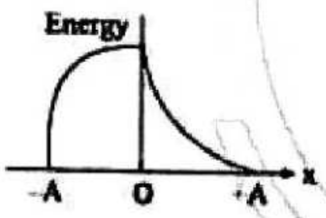
(1) 5:7:3 (2) 5:7:2
(3) 3:7:2 (4) 3:5:2

29. Which graph represents the difference between total energy and potential energy of a particle executing SHM versus it's distance from mean position?

(1)


(2)


(3)


(4)


30. In a linear Simple Harmonic Motion (SHM)
   A. Restoring force is directly proportional to the displacement.
   B. The acceleration and displacement are opposite in direction.
   C. The velocity is maximum at mean position.
   D. The acceleration is minimum at extreme points.

Choose the most appropriate answer from the options given below.
(1) A, B and C only
(2) A, B and D only
(3) A, C and D only
(4) C and D only

31. The function of time representing a simple harmonic motion with a period of π/ω is
(1) $Y = A\sin(\omega t)$ (2) $Y = A\cos(\omega t)$
(3) $Y = A\sin^2(\omega t)$ (4) $Y = A\sin\left(\omega t - \frac{\pi}{4}\right)$

32. For what value of displacement the kinetic energy and potential energy of a simple harmonic oscillation become equal?
(1) $x = 0$ (2) $x = \pm A$
(3) $x = \pm\frac{A}{\sqrt{2}}$ (4) $x = \frac{A}{2}$

33. Two simple harmonic motion, are represented by the equations $y_1 = 10\sin\left(3\pi t + \frac{\pi}{3}\right)$
and $y_2 = 5\left(\sin 3\pi t + \sqrt{3}\cos 3\pi t\right)$
Ratio of amplitude of y_1 to $y_2 = x : 1$. The value of x is ....
(1) 1 : 1 (2) 1 : 2
(3) 1 : 3 (4) 4 : 1

34. Two metallic blocks M_1 and M_2 of same area of cross-section are connected to each other (as shown in figure). If the thermal conductivity of M_2 is K, then the thermal conductivity of M_1 will be : [Assume steady state heat conduction]

(1) 10 K (2) 8 K
(3) 12.5 K (4) 2 K

35. A transverse wave is represented by $y = 2\sin(\omega t - kx)$ cm. The value of wavelength (in cm) for which the wave velocity becomes equal to the maximum particle velocity, will be
(1) 4π (2) 2π
(3) π (4) 2

Physics : Section-B (Q. No. 36 to 50)

36. The equation of a wave on a string of linear mass density 0.04 kg m^{-1} is given by

$$y = 0.02(m)\sin\left[2\pi\left(\frac{t}{0.04(s)} - \frac{x}{0.50(m)}\right)\right]$$

The tension in the string is
(1) 4.0 N (2) 12.5 N
(3) 0.5 N (4) 6.25 N

37. A wave travelling along the x-axis is described by the equation $y(x, t) = 0.005\cos(\alpha x - \beta t)$. If the wavelength and the time period of the wave are 0.08 m and 2.0 s respectively, then α and β in appropriate units are
(1) $\alpha = 25.00\pi, \beta = \pi$
(2) $\alpha = \frac{0.08}{\pi}, \beta = \frac{2.0}{\pi}$
(3) $\alpha = \frac{0.04}{\pi}, \beta = \frac{1.0}{\pi}$
(4) $\alpha = 12.50\pi,\ \beta = \frac{\pi}{2.0}$

38. A travelling wave represented by $y_1 = A\sin(\omega t - kx)$ is superimposed on another wave represented by $y_2 = A\sin(\omega t + kx)$. The resultant is
(1) a standing wave having nodes at $x = \left(n + \frac{1}{2}\right)\frac{\lambda}{2}, n = 0,1,2$
(2) a wave travelling along $+x$ direction
(3) a wave travelling along $-x$ direction
(4) a standing wave having nodes at $x = \frac{n\lambda}{2}$; $n = 0,1,2$

39. An ideal fluid flows (laminar flow) through a pipe of non–uniform diameter. The maximum and minimum diameters of the pipes are 6.4 cm and 4.8 cm, respectively. The ratio of the minimum and the maximum velocities of fluid in this pipe is
(1) 9/16 (2) 81/256
(3) $\sqrt{3}/2$ (4) 3/4

40. The terminal velocity V_t of the spherical rain drop depends on the radius (r) of the spherical rain drop as:
(1) $r^{1/2}$ (2) r
(3) r^2 (4) r^3

41. The bulk modulus of a spherical objects is 'B'. If it is subjected to uniform pressure 'P', the fractional decrease in radius is:
(1) $\frac{B}{3p}$ (2) $\frac{3P}{B}$
(3) $\frac{P}{3B}$ (4) $\frac{P}{B}$

42. If a soap bubble expands, the pressure inside the bubble:
(1) is equal to the atmospheric pressure
(2) decreases
(3) increases
(4) remains the same

43. Match List-I with List-II.

List-I		List-II	
(A)	Isothermal	(I)	Pressure constant
(B)	Isochoric	(II)	Temperature constant
(C)	Adiabatic	(III)	Volume constant
(D)	Isobaric	(IV)	Heat content is constant

Choose the correct answer from the options given below:
(1) A-IV, B-III, C-II, D-I
(2) A-I, B-II, C-IV, D-III
(3) A-III, B-II, C-I, D-IV
(4) A-II, B-III, C-IV D-I

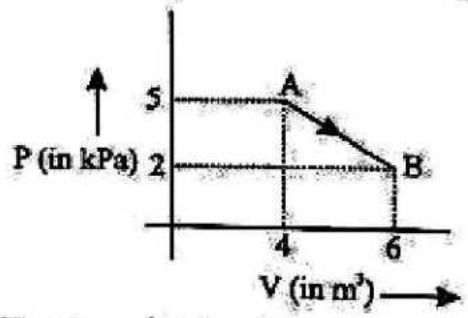
44. The quantities of heat required to raise the temperature of two solid copper spheres of radii r_1 and r_2($r_1 = 1.5\ r_2$) through 1 K are in the ratio:
(1) $\frac{9}{4}$ (2) $\frac{3}{2}$
(3) $\frac{5}{3}$ (4) $\frac{27}{8}$

45. The bulk modulus of a liquid is 3×10^{10} Nm^{-2}. The pressure required to reduce the volume of liquid by 2% is:
(1) 3×10^{8} Nm^{-2} (2) 9×10^{8} Nm^{-2}
(3) 6×10^{8} Nm^{-2} (4) 12×10^{8} Nm^{-2}

46. A cylindrical rod having temperature T_1 and T_2 at its end. The rate of flow of heat Q_1 cal/sec . If all the linear dimension are doubled keeping temperature constant, then rate of flow of heat Q_2 will be:
(1) $4Q_1$ (2) $2Q_1$
(3) $\frac{Q_1}{4}$ (4) $\frac{Q_1}{2}$

47. A wire of length L area of cross section A is hanging from a fixed support. The length of the wire changes to L_1 when mass M is suspended from its free end. The expression for Young's modulus is:
(1) $\frac{Mg(L_1 - L)}{AL}$ (2) $\frac{MgL}{AL_1}$
(3) $\frac{MgL}{A(L_1 - L)}$ (4) $\frac{MgL_1}{AL}$

48. One mole of an ideal diatomic gas undergoes a transition from A to B along a path AB as shown in the figure.



The change in internal energy of the gas during the transition is:
(1) –20 kJ (2) 20 J
(3) –12 kJ (4) 20 kJ

49. One mole of an ideal monoatomic gas undergoes a process described by the equation PV^3 = constant. The heat capacity of the gas during this process is:
(1) $2R$ (2) R
(3) $\frac{3}{2}R$ (4) $\frac{5}{3}R$

50. In a guitar, two strings A and B made of same material are slightly out of tune and produce beats of frequency 6 Hz. When tension in B is slightly decreased, the beat frequency increases to 7 Hz. If the frequency of A is 530 Hz, the original frequency of B will be:
(1) 524 Hz (2) 536 Hz
(3) 537 Hz (4) 523 Hz

Chemistry : Section-A (Q. No. 51 to 85)

51. In the following redox reaction;
$MnO_4^- + SO_3^{-2} + H^+ \longrightarrow SO_4^{2-} + Mn^{2+} + H_2O$
(1) MnO_4^- and H^+ both are reduced
(2) MnO_4^- is reduced and H^+ is oxidised
(3) MnO_4^- is reduced and SO_3^{2-} is oxidised
(4) MnO_4^- is oxidised and SO_3^{2-} is reduced

52. Process of separation of mixture into their components and to purify the compounds by using adsorption is known as;
(1) Sublimation
(2) Differential extraction
(3) Distillation
(4) Chromatography

53. The compound with the highest boiling point is;
(1) n-hexane
(2) n-pentane
(3) 2,2-dimethyl propane
(4) propane

54. Which of the following reactions is used for locating the position of double bond in an alkene?
(1) Hydroboration
(2) Hydroxylation
(3) Chlorohydroxylation
(4) Ozonolysis

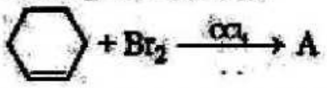
55. Which of the following reactions do not involve redox reaction?
(1) $2Rb + 2H_2O \longrightarrow 2RbOH + H_2$
(2) $2CuI_2 \longrightarrow 2CuI + I_2$
(3) $NH_4Cl + NaOH \longrightarrow NaCl + NH_3 + H_2O$
(4) $3Mg + N_2 \longrightarrow Mg_3N_2$

56. Kolbe electrolytic synthesis can be used to obtain all hydrocarbons as a major product except;
(1) 2-Butene (2) Ethyne
(3) Methane (4) Ethene

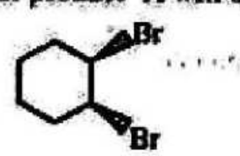
57. In Duma's method for the estimation of nitrogen, 0.84 g of an organic compound gave 448 mL of nitrogen at STP the percentage of nitrogen in the compound is;
(1) 33.3 %
(2) 66.7 %
(3) 50 %
(4) 60 %

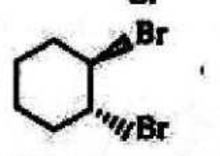
58. Reaction of HBr with propene in the presence of peroxide gives;
(1) 3-bromo propane
(2) Allyl bromide
(3) n-propyl bromide
(4) Isopropyl bromide

59. In the given reaction:



Major product 'A will be;

(1) 

(2) 

(3) Both (1) and (2)
(4) None of these

60. Which of the following statements is incorrect?
(1) In redox systems, the titration method can be adopted to determine the strength of a reductant/oxidant using a redox sensitive indicator.
(2) MnO_4^- acts as a self indicator in the titration.
(3) $Cr_2O_7^{2-}$ acts as a self indicator in the titration.
(4) Iodine itself gives an intense blue colour with starch and has a very specific reaction with thiosulphate ions ($S_2O_3^{2-}$).

61. What is the basis for process of distillation?
(1) Difference in melting point
(2) Difference in temperature
(3) Difference in pressure
(4) Difference in boiling point

62. Which amongst the following is incorrect statement ?
(1) –OH group activates the benzene ring for the attack of an electrophile at o and p- positions.
(2) Halogens are moderately deactivating group in the case of aryl halides.
(3) Chlorobenzene is more reactive than benzene towards the electrophilic substitution reaction.
(4) $-NO_2$ is a meta directing group.

63. What is hybridisation of each carbon atom in following compound from left to right?
$HC\equiv C-CH=CH-CH_3$
(1) sp, sp^2, sp^2, sp^2, sp^3
(2) sp, sp, sp^2, sp^2, sp^3
(3) sp, sp, sp^2, sp^3, sp^3
(4) sp, sp^2, sp^2, sp^3, sp^3

64. Which of the following statement is incorrect?
(1) In a disproportionation reaction, an element in one oxidation state is simultaneously oxidised and reduced.
(2) ClO_4^- undergoes disproportionation reaction.
(3) Cl, Br, & I show disproportionation but fluorine doesn't show.
(4) Phosphorous, sulphur and chlorine undergo dispropor-tionation in the alkaline medium.

65. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: $KMnO_4$, $KClO_4$, and HNO_3 act as oxidants.
Reason R: In $KMnO_4$, $KClO_4$, and HNO_3 central atom is in its lowest oxidation state.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) **A** is true but **R** is false.
(2) **A** is false but **R** is true.
(3) Both **A** and **R** are true and **R** is the correct explanation of **A**.
(4) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

66. Match **List–I** with **List–II**.

List–I (Oxidation)	List–II (Catalyst/Reagent)
(A) $CH_4 + O_2 \rightarrow CH_3OH$	(I) Mo_2O_3 / Δ
(B) $CH_4 + O_2 \rightarrow HCHO$	(II) $(CH_3COO)_2Mn / \Delta$
(C) $CH_3-CH_3 \rightarrow CH_3COOH$	(III) $KMnO_4$
(D) $(CH_3)_3CH \rightarrow (CH_3)_3COH$	(IV) Cu/523 K/100 atm

Choose the **correct** answer from the options given below.
(1) A–IV, B–I, C–II, D–III
(2) A–III, B–IV, C–I, D–II
(3) A–II, B–III, C–I, D–IV
(4) A–I, B–II, C–IV, D–III

67. The third member of the homologous series of aliphatic aldehydes has the structure;
(1) CH_3CH_2CHO (2) $CH_3(CH_2)_2CHO$
(3) $CH_3COCH_2CH_3$ (4) CH_3COCH_3

68. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A: Chromatography involves separation of mixture over a column of adsorption.
Reason R: Higher the value of R_f means higher adsorption.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) **A** is true but **R** is false.
(2) **A** is false but **R** is true.
(3) Both **A** and **R** are true and **R** is the correct explanation of **A**.
(4) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

69. IUPAC name of the compound is;
$$\begin{array}{l} CH_3-CH_2-\underset{\substack{|\\CH_2\\|\\CH_3}}{CH}-\underset{\substack{|\\CH_3}}{CH}-CH_3 \end{array}$$
(1) 3-Ethyl-2-methylpentane
(2) 2-Methyl-3-ethylpentane
(3) 3-Isopropylpentane
(4) 3-(1-Methyl ethyl)pentane

70. Match **List–I** with **List–II**.

List–I (Chemical Species)	List–II (Oxidation no. of central atom)
(A) CrO_5	(I) 0
(B) I_3^-	(II) −1
(C) O_3	(III) +6
(D) F^-	(IV) $-\frac{1}{3}$

Choose the **correct** answer from the options given below.
(1) A–I, B–II, C–III, D–IV
(2) A–III, B–IV, C–I, D–II
(3) A–II, B–III, C–I, D–IV
(4) A–I, B–II, C–IV, D–III

71. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Having same molecular formula aldehydes and ketones are functional isomers to each other.
Reason R: Aldehyde and ketone having same molecular formula but different functional group are functional isomers.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) **A** is true but **R** is false.
(2) **A** is false but **R** is true.
(3) Both **A** and **R** are true and **R** is the correct explanation of A.
(4) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

72. How many moles of $K_2Cr_2O_7$ can be reduced by 1 mole of Sn^{2+}?
(1) $\frac{1}{3}$ (2) $\frac{1}{6}$
(3) $\frac{2}{3}$ (4) 1

73. In the kjeldahl method for estimation of nitrogen present in soil sample, ammonia evolved from 0.75 g of sample neutralised 10 ml of 1 M H_2SO_4. The % of nitrogen in the soil approximately is;
(1) 37 % (2) 45 %
(3) 25 % (4) 60 %

74. Given below are two statements:
Statement I: Rate of reaction of alkanes with halogens is $F_2 > Cl_2 > Br_2 > I_2$.
Statement II: Halogenation of alkane follow free radical mechanism.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement-I is correct but Statement-II is incorrect.
(2) Statement-I is incorrect but Statement-II is correct.
(3) Both Statement-I and Statement-II are correct.
(4) Both Statement-I and Statement-II are incorrect.

75. The following reaction represents;
$Mg + CuO \rightarrow MgO + Cu$
(1) decomposition reaction
(2) combination reaction
(3) displacement reaction
(4) disproportionate reaction

76. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: NO_2 group present on the benzene ring is meta director.
Reason R: NO_2 group direct the incoming electrophile group to meta position, in benzene ring.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) **A** is true but **R** is false.
(2) **A** is false but **R** is true.
(3) Both **A** and **R** are true and **R** is the correct explanation of A.
(4) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

77. Identify **correct** statements regarding electrophile:
A. Electrophile is electron rich reaction intermediate.
B. Electrophile attack at electron rich site present in the substrate.
C. Ethyl carbocation is an electrophile.
Choose the *most appropriate* answer from the options given below:
(1) A, B and C only
(2) A and B only
(3) B and C only
(4) B only

78. Match **List–I** with **List–II**.

	List–I Precipitate		List–II Colour
(A)	PbS	(I)	Violet
(B)	$[Fe(CN)_5NOS]^{4-}$	(II)	Blood red
(C)	$(NH_4)_3PO_4 12MoO_3$	(III)	Black
(D)	$[Fe(SCN)]^{2+}$	(IV)	Yellow

Choose the **correct** answer from the options given below.
(1) A–II, B–III, C–IV, D–II
(2) A–III, B–IV, C–I, D–II
(3) A–II, B–III, C–I, D–IV
(4) A–III, B–I, C–IV, D–II

79. CH_3CH_2–Cl undergoes homolytic fission to produce;

(1) $CH_3\dot{C}H_2$ and $\dot{Cl}$

(2) $CH_3\dot{C}H_2$ and Cl

(3) $CH_3\overset{\oplus}{C}H_2$ and Cl^-

(4) $CH_3\overset{\ominus}{C}H_2$ and $Cl^{\oplus}$

80. Given below are two statements:

Statement I: A negative E° means that the redox couple is a stronger reducing agent than the H^+/H_2 couple.

Statement II: A positive E° mean that the redox couple is a weaker reducing agent than the H^+/H_2 couple.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both Statement I and Statement II are incorrect.

1. Given below are two statements:

Statement I: Electromeric effect is temporary effect.

Statement II: Organic compounds having a multiple bond can show electromeric effect in presence of attacking reagent only.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

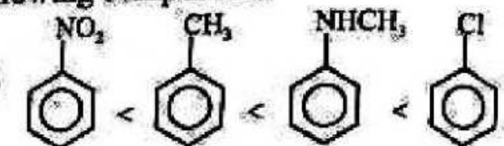
(4) Both Statement I and Statement II are incorrect.

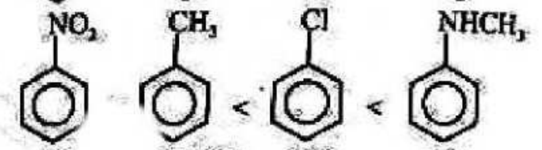
82. In the reaction: $A^{-n_2} + xe^- \rightarrow A^{-n_1}$, here X will be;

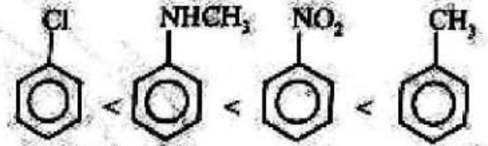
(1) $n_1 + n_2$ (2) $n_2 - n_1$

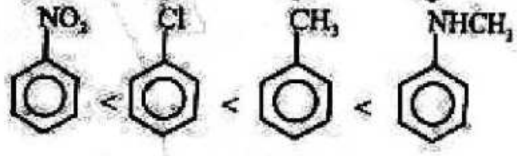
(3) $n_1 - n_2$ (4) $n_1 \times n_2$

83. Identify **correct** order of reactivity for electrophilic substitution reaction of the following compounds:

(1) 

(2) 

(3) 

(4) 

84. Impure sample of naphthalene can be purified by

(1) Distillation (2) Chromatography

(3) Crystallisation (4) Sublimation

85. Match **List–I** with **List–II**.

List–I Carbocation		List–II No. of Hyper-conjugative structures	
(A)	$CH_3-CH_2-\overset{\oplus}{C}H-CH_3$	(I)	0
(B)	$CH_3-\overset{\oplus}{C}H-CH_3$	(II)	3
(C)	$C_6H_5-\overset{\oplus}{C}H_2$	(III)	5
(D)	$CH_3-\overset{\oplus}{C}H_2$	(IV)	6

Choose the **correct** answer from the options given below.

(1) A–III, B–I, C–II, D–IV

(2) A–II, B–III, C–IV, D–I

(3) A–III, B–IV, C–I, D–II

(4) A–I, B–IV, C–II, D–III

Chemistry : Section-B (Q. No. 86 to 100)

86. Consider the following reaction,

$$K_2Cr_2O_7 + 4H_2SO_4 + xH_2O_2 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + yO_2 + 7H_2O$$

x and y are;

(1) $x = 5, y = 2$
(2) $x = 3, y = 3$
(3) $x = 4, y = 10$
(4) $x = 5, y = 5$

87. Condition of aromaticity is/are;

A. System must be planar.
B. Complete delocalisation of the π-electrons in the closed system.
C. Presence of $(4\pi + 2)\pi$ electrons in the ring where n is an integer $(n = 0, 1, 2, ...)$.

Which of the above statements are correct?

(1) A and B only
(2) A and C only
(3) A, B and C only
(4) B and C only

88. +I effect is shown by:

(1) $-NO_2$
(2) $-Cl$
(3) $-Br$
(4) $-CH_3$

89. Qualitative analysis of carbon and hydrogen done by the following method;

A. Carbon present in organic compound detected by lime water.
B. Hydrogen present in organic compound detected by anhydrous copper sulphate.
C. Anhydrous copper sulphate turn black when water is passed through it.

Choose the most appropriate answer from the given options below:

(1) A, B and C only
(2) A and B only
(3) A and C only
(4) B and C only

90. Which of the following is **not** involved in resonance?

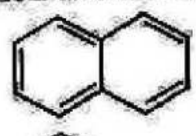
(1) $\overset{\ominus}{C}H_2 - C \equiv N$
(2) $\overset{\oplus}{C}H_2 - CH = O$
(3) $CH_2 = CH - \overset{\oplus}{C}H_2$
(4) $CH_2 = CH - \overset{\oplus}{N}H_3$

91. What is 'A' in the following reaction?

$2Fe^{3+}_{(aq)} + Sn^{2+}_{(aq)} \rightarrow 2Fe^{2+}_{(aq)} + A$

(1) $Sn^{3+}_{(aq)}$
(2) $Sn^{4+}_{(aq)}$
(3) $Sn^{2+}_{(aq)}$
(4) Sn

92. Which of the following is non-aromatic?

(1) 
(2) 
(3) 
(4) 

93. $3CH \equiv CH \xrightarrow[873K]{\text{Red hot iron tube}} A$

The product formed A is;

(1) 
(2) 
(3) 
(4) 

94. In the detection of nitrogen, the sodium fusi extract is acidified with which of the following

(1) dil. H_2SO_4
(2) dil. HCl
(3) conc. H_2SO_4
(4) conc. HCl

95. Which of the following has minimum heat hydrogenation?

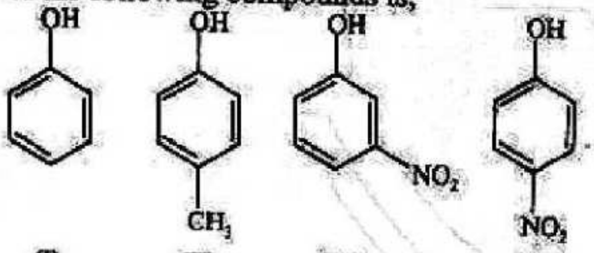
(1) ethene
(2) Propene
(3) 1-butene
(4) cis-2-butene

96. In Br_3O_8 compound, oxidation number of bromine is;
(1) $+\frac{16}{13}$ (2) $+\frac{26}{3}$
(3) $+\frac{24}{3}$ (4) $+\frac{16}{3}$

97. The intermediate during the addition of HCl to propene in the presence of peroxide is;
(1) $CH_3-\dot{C}H-CH_2Cl$
(2) $CH_3-\overset{\oplus}{C}H-CH_3$
(3) $CH_3-CH_2-\overset{\oplus}{C}H_2$
(4) $CH_3-\dot{C}H-CH_3$

98. Which gives Lassaigne's test for nitrogen?
(1) Pyridine
(2) Benzene diazonium chloride
(3) Hydrazine
(4) Hydrazoic acid

99. The correct order of decreasing acidic strength of the following compounds is;



(I) (II) (III) (IV)
(1) III > IV > I > II
(2) I > IV > III > II
(3) II > I > III > IV
(4) IV > III > I > II

100. Which of the following is a example of decomposition reaction?
(1) $2H_2O(\ell) \xrightarrow{\Delta} 2H_2(g) + O_2(g)$
(2) $3H_2(g) + \underset{\text{Nitrogen}}{N_2(g)} \rightarrow \underset{\text{Ammonia}}{2NH_3(g)}$
(3) $\underset{\text{Magnesium oxide}}{MgO(s)} + \underset{\text{Carbon dioxide}}{CO_2(g)} \rightarrow \underset{\text{Magnesium carbonate}}{MgCO_3(s)}$
(4) $\underset{\text{Carbon monoxide}}{2CO(g)} + \underset{\text{Oxygen}}{O_2(g)} \rightarrow \underset{\text{Carbon dioxide}}{2CO_2(g)}$

Botany : Section-A (Q. No. 101 to 135)

101. Who observed that all plants are composed of different kinds of cells?
(1) Schleiden
(2) Schwann
(3) Rudolf Virchow
(4) Robert Brown

102. Find the **mismatched** pair:
(1) Metaphase I - Bivalent chromosomes align on the equatorial plate
(2) Prophase II- The nuclear membrane disappears by the end of prophase II
(3) Anaphase I- Sister chromatids separate
(4) Telophase II- Meiosis ends with telophase II

103. The organelle which is **not** a part of the endomembrane system is;
(1) ER.
(2) Golgi complex.
(3) lysosomes.
(4) mitochondria.

104. Photorespiration is a wasteful process because there is;
(1) neither synthesis of phosphoglycerate, nor of ATP.
(2) neither synthesis of phosphoglycerate, nor of phosphoglycolate.
(3) neither synthesis of sugars, nor of ATP.
(4) neither synthesis of phosphoglycolate nor of ATP.

105. The formation of which of the following is the example of dedifferentiation?
(1) Procambium and vascular cambium
(2) Cork cambium and interfascicular cambium
(3) Cork cambium and vascular cambium
(4) Procambium and cork cambium

106. In glycolysis, glucose undergoes partial oxidation to form;
(1) one molecule of pyruvic acid.
(2) two molecules of pyruvic acid.
(3) three molecules of pyruvic acid.
(4) four molecules of pyruvic acid.

107. Given below are two statements: one is labeled as Assertion A and the other is labelled as Reason R:
Assertion A: Prophase is marked by the initiation of condensation of chromosomal material.
Reason R: The centrosome, which had undergone duplication during S phase of interphase, now begins to move towards centre of the cell.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) **A** is true but **R** is false.
(2) **A** is false but **R** is true.
(3) Both **A** and **R** are true and **R** is the correct explanation of **A**.
(4) Both **A** and **R** are true but **R** is **NOT** the correct explanation of **A**.

108. Read the following four statements, (i), (ii), (iii) and (iv) and select the **correct** statements.
(i) Z scheme of light reaction takes place in presence of PS I only.
(ii) Only PS I is functional in cyclic photophosphorylation.
(iii) Cyclic photophosphorylation results in synthesis of ATP and $NADPH_2$.
(iv) Stroma lamellae lack PS II as well as NADP reductase.
(1) (ii) and (iv) (2) (i) and (ii)
(3) (ii) and (iii) (4) (iii) and (iv)

109. In both lactic acid and alcohol fermentation, how much of the energy in glucose is released?
(1) More than 12 percent
(2) More than 7 percent
(3) Less than 7 percent
(4) More than 50 percent

110. Which of the following may be traversed by plasmodesmata?
(1) Cell wall & nucleus.
(2) Cell wall & glycocalyx.
(3) Cell wall only.
(4) Cell wall & middle lamella.

111. Choose the **incorrect** statement:
(1) Meiosis involves two sequential cycles of nuclear and cell division called meiosis I and meiosis II but only a single cycle of DNA replication.
(2) Meiosis I is initiated after the parental chromosomes have replicated to produce identical sister chromatids at the S phase.
(3) Meiosis involves pairing of homologous chromosomes and recombination between sister chromatids of homologous chromosomes.
(4) Four haploid cells are formed at the end of meiosis II.

112. In animal cells lipid-like steroidal hormones are synthesised in;
(1) rough endoplasmic reticulum.
(2) Golgi bodies.
(3) lysosomes.
(4) smooth endoplasmic reticulum.

113. Read the given statements.
Statement I: In metaphase, spindle fibres attach to kinetochores of chromosomes.
Statement II: In anaphase, centromeres split and chromatids separate.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Statement-I is correct but Statement-II is incorrect.
(2) Statement-I is incorrect but Statement-II is correct.
(3) Both Statement-I and Statement-II are correct.
(4) Both Statement-I and Statement-II are incorrect.

114. Arrange the following steps in the correct order as they occur in the C_4 photosynthesis pathway:
(a) C_4 acid transport from mesophyll to bundle sheath cells
(b) Decarboxylation of C_4 acid
(c) CO_2 fixation in mesophyll cells
(d) Regeneration of the PEP in mesophyll cells
Choose the correct sequence from the options given below:
(1) (c)-(a)-(b)-(d) (2) (a)-(c)-(b)-(d)
(3) (c)-(d)-(a)-(b) (4) (d)-(c)-(a)-(b)

115. Choose the **incorrect** statement:
(1) The prokaryotic cells are represented by bacteria, blue-green algae, mycoplasma and PPLO (Pleuro Pneumonia Like Organisms).
(2) Prokaryotic cells are generally smaller and multiply more rapidly than the eukaryotic cells.
(3) All prokaryotes have a cell wall surrounding the cell membrane except in mycoplasma.
(4) No organelles, like the ones in eukaryotes, are found in prokaryotic cells except for the nucleus.

116. Match the **List-I** and **List-II.**

List-I		List-II
(A) Metacentric	(I)	centromere is slightly away from the middle of the chromosome resulting in one shorter arm and one longer arm.
(B) Sub-metacentric	(II)	chromosomes have a terminal centromere.
(C) Acrocentric	(III)	chromosome has middle centromere forming two equal arms of the chromosome
(D) Telocentric	(IV)	centromere is situated close to its end forming one extremely short and one very long arm.

Choose the **correct** answer from the options given below:
(1) A-II, B-IV, C-I, D-III
(2) A-I, B-II, C-III, D-IV
(3) A-III, B-I, C-IV, D-II
(4) A-IV, B-III, C-II, D-I

117. Which of the following statement(s) is/are **true** about Blackman's Law of Limiting Factors?
A. If a physical process is affected by more than one factor, then its rate will be determined by the factor which is nearest to its minimal value.
B. If a chemical process is affected by more than one factor, then its rate will be determined by the factor which is nearest to its minimal value.
C. It is the factor which indirectly affects the process if its quantity is changed.
D. Even with light, CO_2, and a green leaf, photosynthesis won't occur in very low temperatures but resumes at optimal temperatures.

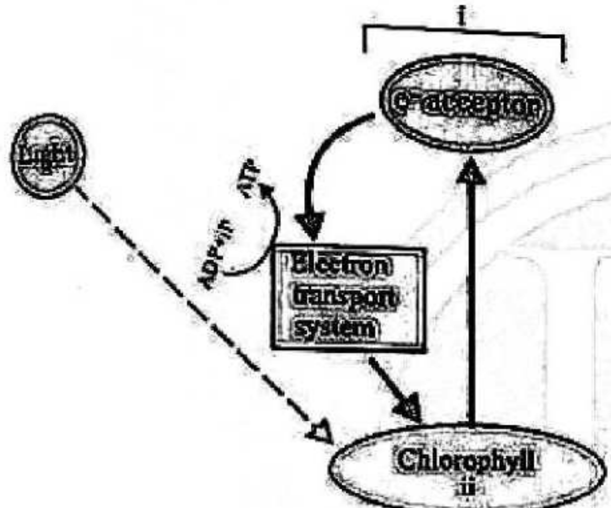
Choose the *most appropriate* answer from the options given below:
(1) (A) and (B)
(2) (B) and (D)
(3) (A), (B) and (C)
(4) All of these

118. An organism has 12 chromosomes (2n) in each cell. During the interphase of mitosis if the number of chromosomes at G_1 phase is 12, what would be the number of chromosomes after S phase?
(1) 12
(2) 24
(3) 36
(4) 6

119. Given below are two statements:
Statement-I: The final product of glycolysis, pyruvate is transported from the cytoplasm into the mitochondria.
Statement-II: The TCA cycle starts with the condensation of acetyl group with oxaloacetic acid (OAA) and water to yield citric acid.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Statement-I is correct but Statement-II is incorrect.
(2) Statement-I is incorrect but Statement-II is correct.
(3) Both Statement-I and Statement-II are correct.
(4) Both Statement-I and Statement-II are incorrect.

120. Find the **mismatch** pair w.r.t the plant growth regulators.
   (1) Adenine derivative- kinetin
   (2) Carotenoid derivative - abscisic acid
   (3) Terpene derivative- gibberellic acid
   (4) Gaseous hormone- indole 3-acetic acid

121. Identify the labeled part in cyclic photophosphorylation.



   (1) i-photosystem II, ii-P 700
   (2) i-photosystem I, ii-P 680
   (3) i-photosystem I, ii-P 700
   (4) i-photosystem II, ii-P 680

122. Which of the following occurs during zygotene phase of meiosis?
   (1) Spindle fibres
   (2) Disappearance of nucleolus
   (3) Chromosomes movement
   (4) Synapsis

123. Find the **incorrectly** matched pair.
   (1) Yeast - cell cycle in only about 90 minutes.
   (2) M Phase - the actual cell division or mitosis occurs.
   (3) Interphase - lasts not more than 45% of the duration of cell cycle.
   (4) Karyokinesis - separation of daughter chromosomes.

124. Match the following.

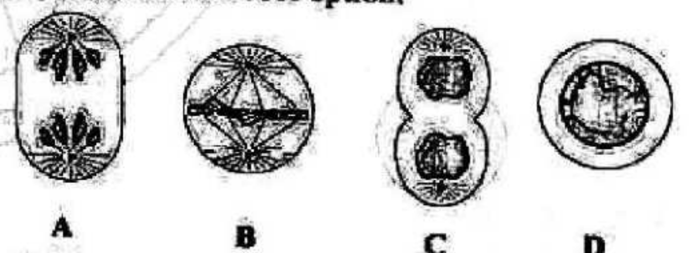
	Column-I		Column-II
(a)	IAA	(i)	Herring sperm DNA
(b)	ABA	(ii)	Bolting
(c)	Ethylene	(iii)	Stomatal closure
(d)	GA	(iv)	Weed-free lawns
(e)	Cytokinins	(v)	Ripening of fruits

   (1) (a)-(iv) (b)-(iii) (c)-(v) (d)-(ii) e-(i)
   (2) (a)-(v) (b)-(iii) (c)-(iv) (d)-(ii) e-(i)
   (3) (a)-(v) (b)-(i) (c)-(iv) (d)-(iii) e-(ii)
   (4) (a)-(v) (b)-(iii) (c)-(ii) (d)-(i) e-(iv)

125. Read the following statements and choose the set of **incorrect** statements.
   A. Golgi apparatus is the important site of formation of steroidal hormones.
   B. The mitochondria divide by fission.
   C. The leucoplasts are the colourless plastids.
   D. The vacuole is found in the nucleoplasm of plant cell.

   (1) A and C (2) A and D
   (3) B and C (4) B and D

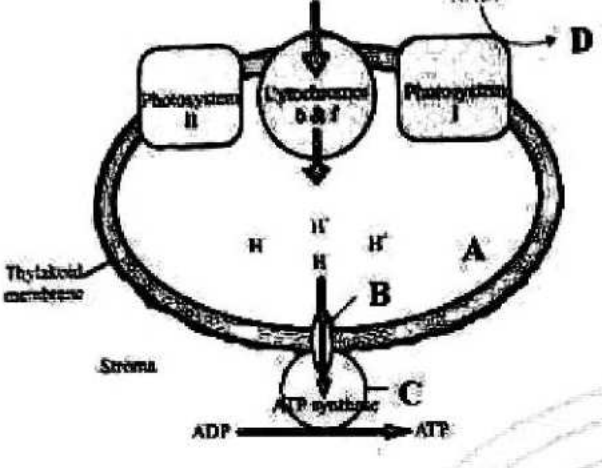
126. Identify the given stages of mitosis given below and select the **correct** option:



A B C D

   (1) A – Anaphase, B – Metaphase, C – Telophase, D – Prophase.
   (2) A – Metaphase, B – Telophase, C – Anaphase, D – Prophase.
   (3) A – Telophase, B – Metaphase, C – Prophase, D – Anaphase.
   (4) A – Telophase, B – Metaphase, C – Anaphase, D – Prophase.

127. Identify the parts marked as A, B, C and D in the given figure showing ATP synthesis through chemiosmosis.



(1) A-Thylakoid lumen, B- CF_0, C- CF_1, D-NADPH
(2) A- Thylakoid lumen, B- CF_1, C-CF_0, D-NAD
(3) A- Chloroplast lumen, B- CF_0, C- CF_1, D-FADH
(4) A- Chloroplast lumen, B- CF_1, C-CF_0, D-NAD

128. Arrange the following events of meiosis in correct sequence and choose the **correct** option:
A. Crossing over
B. Synapsis
C. Terminalisation of chiasmata
D. Disappearance of nucleolus
(1) B, A, C, D (2) A, B, C, D
(3) B, C, D, A (4) B, A, D, C

129. The developmental stages in a plant are as follows:
(1) Division → Differentiation → Maturation → Senescence → Death
(2) Division → Maturation → Differentiation → Expansion → Death
(3) Division → Differentiation → Maturation → Expansion → Senescence → Death
(4) Division → Expansion → Differentiation → Maturation → Senescence → Death

130. Given below are two statement: one is labeled as Assertion A and the other is labelled as Reason R:
Assertion A: Larger and more numerous nucleoli are present in cells actively carrying out protein synthesis.
Reason R: Nucleolus is a site for active ribosomal RNA synthesis.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) **A** is true but **R** is false.
(2) **A** is false but **R** is true.
(3) Both **A** and **R** are true and **R** is the correct explanation of **A**.
(4) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

131. Match the List-I and List-II.

List-I		List-II	
(A)	Amyloplasts	(I)	Store proteins
(B)	Chromoplasts	(II)	Store oil and fats
(C)	Aleuroplasts	(III)	Store carbohydrates
(D)	Elaioplasts	(IV)	Carotenoid pigments

Choose the **correct** answer from the options given below:
(1) A-II; B-III; C-IV; D-I
(2) A-III; B-IV; C-I; D-II
(3) A-IV; B-III; C-II; D-I
(4) A-III; B-II; C-I; D-IV

132. Read the facts about cytoplasm:
I. main arena of cellular activities.
II. to keep the cell in the 'living state'.
III. in both prokaryotic and eukaryotic cells.
IV. a semi-fluid matrix.
Choose the **correct** code given below;
(1) I and II only
(2) II and III only
(3) I, II and III only
(4) I, II, III and IV

133. Identify the products generated by the light-dependent reactions of photosynthesis:
(a) Oxygen (b) Carbon dioxide
(c) Glucose (d) ATP
(e) NADPH
Choose the correct answer from the options given below:
(1) (a), (d), and (e) only
(2) (b), (c), and (d) only
(3) (a), (c), and (e) only
(4) (b), (d), and (e) only

134. Choose the **incorrectly** matched pair among the following
(1) Maturation phase - Cells have attained their maximal size in terms of cytoplasmic increase.
(2) Geometric growth - Exponential in nature
(3) Oxygen - Releases metabolic energy
(4) Differentiation - Tracheary elements devoid of protoplasm.

135. How many of the given events are **correctly** related with photorespiration?
I. No synthesis of sugars.
II. No synthesis of ATP and NADPH.
III. Occurs in C_4 plants.
IV. Biological function of photorespiration is known properly.
Choose the *most appropriate* answer from the options given below:
(1) Two (2) Three
(3) Four (4) None of these

Botany : Section-B (Q. No. 136 to 150)

136. The given figure shows:

(1) Developmental heterophylly in buttercup.
(2) Environmental heterophylly in larkspur.
(3) Environmental heterophylly in buttercup.
(4) Developmental heterophylly in larkspur.

137. Match List-I with List-II.

List-I (Location)	List-II (Function/Characteristic)
(A) Stroma of chloroplast	(I) Chemiosmosis
(B) Lumen of the thylakoids	(II) Presence of both PS I and PS II
(C) Lamellae of the grana	(III) Splitting of water
(D) Membranes of thylakoid	(IV) Enzymatic reactions synthesize sugar

Choose the **correct** answer from the options given below:
(1) A-IV, B-III, C-II, D-I
(2) A-I, B-III, C-II, D-IV
(3) A-IV, B-II, C-III, D-I
(4) A-IV, B-I, C-II, D-III

138. Match the columns and find out the correct combination:

Column-I	Column-II
(a) Joseph Priestley	(i) Used a prism
(b) T.W. Engelmann	(ii) Studied purple and green bacteria
(c) Cornelius van Niel	(iii) Performed bell jar experiment

(1) (a)-(i) (b)-(iii) (c)-(ii)
(2) (a)-(iii) (b)-(i) (c)-(ii)
(3) (a)-(ii) (b)-(iii) (c)-(i)
(4) (a)-(i) (b)-(ii) (c)-(iii)

139. Match the columns and find out the correct combination:

Column-I	Column-II
(a) Linear curve	(i) Heterophylly
(b) Redifferentiation	(ii) Secondary xylem
(c) Plasticity	(iii) $L_t = L_0 + rt$
(d) Cork cambium	(iv) Dedifferentiation

(1) (a)-(i) (b)-(iii) (c)-(iv) (d)-(ii)
(2) (a)-(ii) (b)-(iii) (c)-(i) (d)-(iv)
(3) (a)-(iii) (b)-(ii) (c)-(i) (d)-(iv)
(4) (a)-(ii) (b)-(i) (c)-(iii) (d)-(iv)

140. One scientist cultured *Cladophora* in a suspension of *Azotobacter* and illuminated the cultured by splitting light through a prism. He observed that bacteria accumulated mainly in the region of;

(1) violet and green light.
(2) indigo and green light.
(3) orange and yellow light.
(4) blue and red light.

141. When CO_2 is added to PEP, the first stable product synthesized is;

(1) pyruvate.
(2) glyceraldehyde-3-phosphate.
(3) phosphoglycerate.
(4) oxaloacetate.

142. Reduction process of Calvin cycle requires how many ATP and NADPH for the reduction of one molecule of CO_2?

(1) 2 mole ATP and 3 mole NADPH.
(2) 2 mole ATP and 2 mole NADPH.
(3) 1 mole ATP and 2 mole NADPH.
(4) 3 mole ATP and 2 mole NADPH.

143. Find out the **incorrect** statement from the following.

(1) Dark reaction depends on the product formed by light reaction.
(2) In stroma, enzymatic reactions incorporate CO_2 into the plant leading to the synthesis of sugar.
(3) Purple and green sulphur bacteria use H_2S as hydrogen donor.
(4) There is no division of labour in chloroplast.

144. Read the following statements:

(i) There can be a net gain of 38 ATP molecules during aerobic respiration of one molecule of glucose.
(ii) Yeasts poison themselves to death when the concentration of alcohol reaches about 13%.
(iii) In both lactic acid and alcoholic fermentation not much energy is released, less than 7% of the energy in glucose is released and not all of it is trapped as high energy bonds of ATP.
(iv) In fermentation, say by yeast, the complete oxidation of glucose is achieved under aerobic conditions.
(v) During aerobic respiration, complete oxidation of respiratory substrate releases energy, CO_2 but not water.

How many of the above statements are correct?

(1) Five
(2) Four
(3) Three
(4) Two

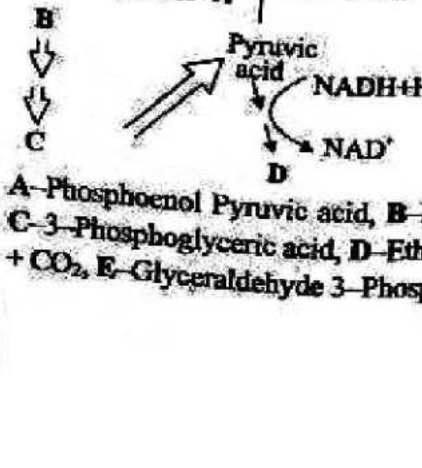
145. Match **List-I** with **List-II**, regarding shapes and size of cells.

List-I	List-II
(A) Largest isolated single cell	(I) egg of an ostrich
(B) Smallest cells	(II) mycoplasmas
(C) Red blood cells	(III) round and biconcave
(D) White blood cells	(IV) amoeboid

Select the **correct** option from the codes given below;

(1) A-I, B-II, C-III, D-IV
(2) A-III, B-IV, C-II, D-I
(3) A-II, B-IV, C-III, D-I
(4) A-IV, B-III, C-II, D-I

146. Identify **A** to **E** in the given figure.



(1) A–Phosphoenol Pyruvic acid, **B**–Lactic acid, **C**–3–Phosphoglyceric acid, **D**–Ethanol + CO_2, **E**–Glyceraldehyde 3–Phosphate.

(2) A–Glyceraldehyde 3-Phosphate, B–3 Phosphoglyceric acid, C–Phosphoenol Pyruvic acid, D–Ethanol + CO_2, E–Lactic acid.
(3) A–Lactic acid, B–Ethanol + CO_2, C–Glyceraldehyde 3-Phosphate, D–Phosphoenol Pyruvic acid, E–3–Phosphoglyceric acid.
(4) A–3-Phosphoglyceric acid, B–Phosphoenol Pyruvic acid, C–Lactic acid, D–Glyceraldehyde 3–Phosphate, E–Ethanol + CO_2

147. Which statement is **wrong** for Krebs' cycle?
(1) There are three points in the cycle where NAD^+ is reduced to $NADH + H^+$.
(2) There is one point in the cycle where FAD^+ is reduced to $FADH_2$.
(3) During conversion of succinyl CoA to succinic acid, a molecule of GTP is synthesized.
(4) The cycle starts with condensation of acetyl group (acetyl CoA) with pyruvic acid to yield citric acid.

148. Read the following statements regarding cell cycle.
A. Sequence by which cell duplicates its genome, synthesises the other constituents and eventually divides into two daughter cells is termed as cell cycle.
B. Cell growth (in terms of cytoplasmic increase) is a continuous process.
C. DNA synthesis occurs only during one specific stage in the cell cycle.
D. Duration of cell cycle can vary from organism to organism and also from cell type to cell type.
Choose the **correct** options.
(1) A and B only
(2) B and C only
(3) A, B and D only
(4) A, B, C and D

149. Select the correctly matched pairs.

	Stages of respiration	Place of occurrence	Products
A.	Glycolysis	Cytoplasm	Pyruvic acid, $NADH_2$, ATP
B.	Oxidative decarboxylation of acid	Mitochondrial matrix	Acetyl CoA, $NADH_2$, CO_2
C.	Krebs cycle	Perimitochondrial space	CO_2, $NADH_2$, ATP
D.	Electron transport system	outer membrane of mitochondria	ATP, CO_2

(1) A and B
(2) C and D
(3) A and C
(4) B and D

150. Given below are two statements.
Statement I: Glycocalyx differs in composition and thickness among different bacteria.
Statement II: Glycocalyx could be a loose sheath called the capsule in some bacteria, while in others it may be thick and tough, called the slime layer.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

Zoology : Section-A (Q. No. 151 to 185)

151. Which of the following hormones is released by neurohypophysis?
(1) Luteinizing hormone
(2) Oxytocin
(3) Aldosterone
(4) Prolactin

152. Given below are some statements **(A-D)**.
A. Initial bronchioles are supported by incomplete cartilaginous rings.
B. Terminal bronchioles give rise to alveoli.
C. The larynx is a bony box and helps in sound production.
D. The epiglottis is a thick cartilaginous flap that covers the glottis during swallowing.
Choose the **incorrect** statement(s) from the options given below:
(1) A, B and D only (2) B and C only
(3) C and D only (4) C only

153. Which of the following types of movements is seen in cells of the human body?
(1) Amoeboid (2) Ciliary
(3) Muscular (4) All of these

154. Match **List-I** with **List-II**:

List-I		List-II
(A) Glycosuria	(I)	Presence of ketone bodies in urine
(B) Ketonuria	(II)	Presence of glucose in urine
(C) Uremia	(III)	Inflammation of glomeruli
(D) Glomerulonephritis	(IV)	Accumulation of urea in blood

Choose the **correct** answer from the options given below:
(1) A-II, B-I, C-IV, D-III
(2) A-I, B-III, C-II, D-IV
(3) A-IV, B-I, C-II, D-III
(4) A-III, B-II, C-IV, D-I

155. On what basis are neurons classified as unipolar, bipolar and multipolar?
(1) On the basis of their sensory nature.
(2) On the basis of their motor nature.
(3) On the basis of number of axon and dendrites.
(4) On the basis of number of cell body.

156. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Closed circulatory system is more advantageous than open circulatory system.
Reason R: The flow of fluid can be more precisely regulated in closed circulatory system.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) **A** is true but **R** is false.
(2) **A** is false but **R** is true.
(3) Both **A** and **R** are true and **R** is the correct explanation of **A**.
(4) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

157. Given below are two statements:
Statement I: The pars distalis produces melanocyte-stimulating hormone (MSH).
Statement II: Melatonin influences our defense mechanism.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

158. In comparison to the solubility of O_2 in blood the solubility of CO_2 is:
(1) 20 - 25 times less.
(2) 2 - 3 times less.
(3) 2 - 3 times higher.
(4) 20 - 25 times higher.

159. Given below are some statements (A-D).
A. About 40-50 per cent of the body weight of a human adult is contributed by muscles.
B. Skeletal muscles are primarily involved in changes of body postures.
C. Activities of smooth muscles are under voluntary control of the nervous system.
D. Activities of cardiac muscles are not in direct control of the nervous system.

Choose the **correct** statements from the options given below:
(1) A, B and D only
(2) B and D only
(3) A, B and C only
(4) A, B, C and D

160. In micturition:
(1) smooth muscles of urinary bladder relax.
(2) urethral sphincter relaxes.
(3) skeletal muscles contracts.
(4) urethral sphincter contracts.

161. Match **List-I** with **List-II**:

List-I		List-II
(A) Cerebral aqueduct	(I)	Outer layer
(B) Dura mater	(II)	Mid brain
(C) Pia mater	(III)	Thin middle layer
(D) Arachnoid mater	(IV)	In contact with brain tissue

Choose the **correct** answer from the options given below:
(1) A-II, B-I, C-IV, D-III
(2) A-I, B-II, C-III, D-IV
(3) A-IV, B-II, C-III, D-I
(4) A-III, B- II, C-IV, D-I

162. If the stroke volume is 100 ml and cardiac output is 10,000 ml/min, find heart rate.
(1) 50 beats/min
(2) 100 beats/min
(3) 200 beats/min
(4) 10 beats/min

163. Arrange the following events in the correct sequence as they occur during the inspiration phase of breathing:
(a) Increase in pulmonary volume
(b) Contraction of diaphragm and external inter-costal muscles
(c) Air moves into the lungs
(d) Increase in thoracic volume
(e) Decrease in intra-pulmonary pressure

Options:
(1) (b), (d), (a), (e), (c)
(2) (b), (e), (d), (a), (c)
(3) (d), (b), (a), (e), (c)
(4) (e), (b), (d), (a), (c)

164. A deep cleft divides the cerebrum longitudinally into two halves, which are connected to each others by tract of nerve fibres called:
(1) corpus luteum.
(2) corpus callosum.
(3) corpora quadrigemina.
(4) cerebral aqueduct.

165. Given below are two statements:
Statement I: The primary function of sweat is to facilitate a cooling effect on the body surface.
Statement II: Our lungs remove small amounts of CO_2 approximately 20 mL/minute.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

166. The role of thymus in humans is chiefly concerned with:
(1) reproduction.
(2) development of the immune system.
(3) calcium balance.
(4) blood coagulation.

167. Given below are two statements:
Statement I: A common collagenous connective tissue layer called fascia, holds the muscle bundles together.
Statement II: Each myofibril has alternate dark and light bands on it.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

168. Cardiac arrest is when:
(1) the heart muscle is suddenly damaged by an inadequate blood supply.
(2) the lungs are congested.
(3) the heart stops beating.
(4) the vessels that supply blood to the heart muscle are affected.

169. Given below are some statements (A-D).
A. CO_2 is carried by haemoglobin as carbamino-haemoglobin, about 50-55 per cent.
B. Plasma contains a very high concentration of an enzyme called carbonic anhydrase.
C. Asthma is due to inflammation of the bronchi and bronchioles.
D. One of the major causes of emphysema is cigarette smoking.
Choose the **correct** statements from the options given below:
(1) A and C only
(2) A and B only
(3) C and D only
(4) B and D only

170. Which statement is **correct** for muscle contraction?
(1) The length of the H-zone is increased.
(2) The length of the A-bands remains constant.
(3) The length of the I-bands gets increased.
(4) The length of two Z-lines gets increased.

171. Match **List-I** with **List-II**:

List-I	List-II
(A) Thyrocalcitonin	(I) Steroid hormone
(B) Aldosterone	(II) Amino acid derivative
(C) Thyroxine	(III) Protein hormone
(D) Adrenaline	(IV) Iodothyronine

Choose the **correct** answer from the options given below:
(1) A-II, B-III, C-IV, D-I
(2) A-II, B-I, C-IV, D-III
(3) A-III, B-I, C-IV, D-II
(4) A-III, B-II, C-IV, D-I

172. Average pH of human urine is:
(1) 6.0.
(2) 9.0.
(3) 3.0.
(4) 14.0.

173. Which of the components of the brain contain centers which control respiration, cardiovascular reflexes and gastric secretions?
(a) Cerebral cortex
(b) Medulla oblongata
(c) Hypothalamus
(d) Cerebellum
Options:
(1) (b) and (c) only
(2) (a) and (d) only
(3) (b) only
(4) (c) and (d) only

174. The trachea is a straight tube extending up to the mid-thoracic cavity, which divides at the level of:
(1) 3rd thoracic vertebrae.
(2) 4th thoracic vertebrae.
(3) 5th thoracic vertebrae.
(4) 7th thoracic vertebrae.

175. Match **List-I** with **List-II**:

List-I		List-II	
(A)	Neutrophils	(I)	Humoral immune response
(B)	Eosinophils	(II)	Inflammatory reactions
(C)	Basophils	(III)	Phagocytosis
(D)	B lymphocytes	(IV)	Allergic reactions

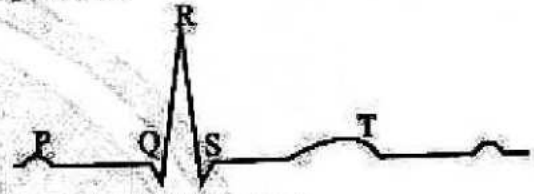
Choose the **correct** answer from the options given below:
(1) A-III, B-IV, C-I, D-II
(2) A-IV, B-III, C-II, D-I
(3) A-III, B-IV, C-II, D-I
(4) A-II, B-IV, C-III, D-I

176. Given below are two statements:
Statement I: Protonephridia are primarily concerned with osmoregulation.
Statement II: Nephridia are the tubular excretory structures of earthworms.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

177. Which of the following hormones is **not** secreted by the gastrointestinal tract?
(1) Gastrin
(2) Secretin
(3) Cholecystokinin
(4) Erythropoietin

178. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The glomerular filtration is considered as a process of selective reabsorption.
Reason R: Blood is filtered so finely through the filtration layers, that almost all the constituents of the plasma except the proteins pass onto the lumen of the Bowman's capsule.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) **A** is true but **R** is false.
(2) **A** is false but **R** is true.
(3) Both **A** and **R** are true and **R** is the correct explanation of **A**.
(4) Both **A** and **R** are true and **R** is **NOT** the correct explanation of **A**.

179. In the given below diagram of ECG, T-wave represents:



(1) atrial depolarisation.
(2) ventricular depolarisation.
(3) ventricular repolarisation.
(4) atrial repolarisation.

180. The inner parts of cerebral hemispheres and a group of associated deep structures like amygdala, hippocampus, etc., form a complex structure called:
(1) thalamus. (2) brainstem.
(3) limbic system. (4) None of these

181. The strength of inspiration and expiration can be increased with the help of additional muscles in the:
(1) ribs. (2) diaphragm.
(3) abdomen. (4) thorax.

182. Given below are some statements **(A-D)**.
A. Progesterone acts on mammary glands and stimulates the formation of alveoli.
B. Several non-endocrine tissues secrete hormones called growth factors.
C. The ovary is composed of ovarian follicles and stromal tissues.

D. Androgens act on the autonomic neural system and influence male sexual behaviour (libido).

Choose the **incorrect** statement(s) from the options given below:

(1) A and D only (2) A and B only
(3) B, C and D only (4) D only

183. A cup-shaped bone called ______ covers the knee ventrally (knee cap).

(1) tibia
(2) fibula
(3) patella
(4) stapes

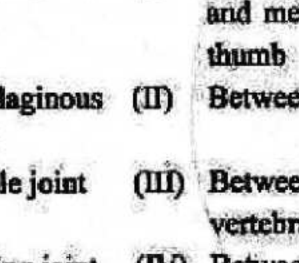
184. Match **List-I** with **List-II:**

List-I		List-II
(A) Fibrous joint	(I)	Between the carpal and metacarpal of the thumb
(B) Cartilaginous joint	(II)	Between the carpals
(C) Saddle joint	(III)	Between adjacent vertebrae
(D) Gliding joint	(IV)	Between skull bones

Choose the **correct answer** from the options given below:

(1) A-IV, B-III, C-I, D-II
(2) A-IV, B-III, C-II, D-I
(3) A-III, B-IV, C-II, D-I
(4) A-II, B-IV, C-III, D-I

185. The given diagram represents the human respiratory system with few structures labelled as **I, II, III and IV**.



The exchange of gases takes place in which of the labelled structures?

(1) I → trachea
(2) II → bronchi
(3) III → bronchioles
(4) IV → alveoli

Zoology : Section-B (Q. No. 186 to 200)

186. Given below are some statements **(A-D).**

A. Atrial natriuretic factor (ANF) mechanism, acts as a check on the renin-angiotensin mechanism.
B. Atrial natriuretic factor converts angiotensin I to angiotensin II in the blood.
C. Angiotensin II is a powerful vasoconstrictor.
D. Aldosterone causes reabsorption of Na^+ and water from the distal parts of the tubule.

Choose the **correct** statements from the options given below:

(1) A and B only (2) B and C only
(3) A, C and D only (4) A, B, C and D

187. Match **List-I** with **List-II:**

List-I		List-II
(A) Earthworm	(I)	Moist cuticle
(B) Mollusca	(II)	Simple diffusion of gases across the body surface
(C) Flatworm	(III)	Gills
(D) Birds	(IV)	Lungs

Choose the **correct answer** from the options given below:

(1) A-IV, B-I, C-II, D-III
(2) A-I, B-III, C-II, D-IV
(3) A-I, B-IV, C-II, D-III
(4) A-IV, B-II, C-III, D-I

188. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

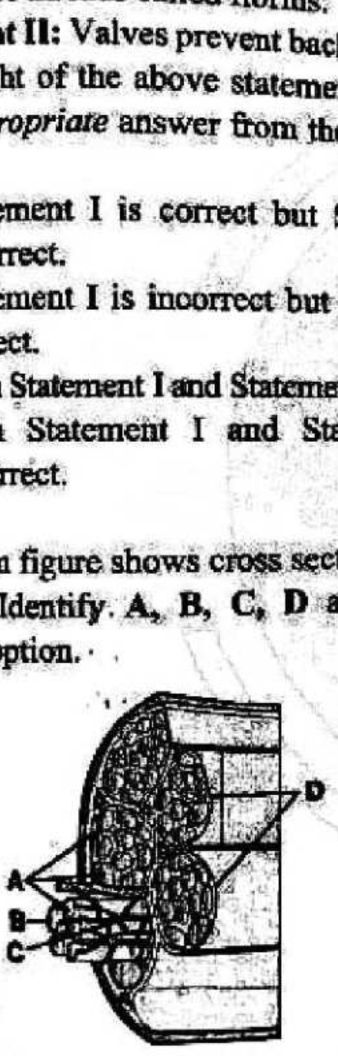
Assertion A: Impulse transmission across an electrical synapse is always slow.

Reason R: At electrical synapses, the membranes of pre and post-synaptic neurons are in very close proximity.

In the light of the above statements, choose the correct answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

189. Given below are two statements:
Statement I: A clot is formed mainly of a network of threads called fibrins.
Statement II: Valves prevent backflow of blood.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

190. The given figure shows cross sectional view of a muscle. Identify. **A, B, C, D** and choose the **correct** option.



(1) A-Myofibrils, B-Sarcoplasmic reticulum, C-Actin filament, D- Fascia
(2) A-Muscle cell, B-Sarcolemma, C-Blood capillary, D-Fascicle
(3) A-Muscle cell, B-Sarcolemma, C-Myosin filament, D-Fascicle
(4) A-Myofibrils, B-Sarcolemma, C-Blood capillary, D-Fascia

191. When a stimulus is applied at a site on the polarised membrane, it becomes freely permeable to ______ ions.
(1) Na^+ (2) K^+
(3) H^+ (4) Ca^{++}

192. Match List-I with List-II:

List-I		List-II	
(A)	Pituitary gland	(I)	Atrial natriuretic factor
(B)	Ovaries	(II)	Follicle-stimulating hormone
(C)	Pancreas	(III)	Estrogen
(D)	Heart	(IV)	Glucagon

Choose the **correct** answer from the options given below:
(1) A-II, B-III, C-IV, D-I
(2) A-III, B-II, C-I, D-IV
(3) A-I, B-II, C-III, D-IV
(4) A-II, B-I, C-III, D-IV

193. Given below are some statements **(A-D)**.
A. The fluid present in the lymphatic system is called the lymph.
B. Lymph has greater mineral distribution than in plasma.
C. Lymph contains specialised lymphocytes.
D. Lymph is an important carrier for nutrients and hormones.
Choose the **incorrect** statement(s) from the options given below:
(1) B only
(2) A and B only
(3) B, C and D only
(4) B and D only

194. After a forced expiration, the capacity of maximum inspiration of the lung is called:
(1) total lung capacity.
(2) functional residual capacity.
(3) vital capacity.
(4) inspiratory capacity.

195. Given below are two statements:
Statement I: Thick filaments in A bands are held together by a thick fibrous membrane called 'M' line.
Statement II: Muscle fibre is a syncitium as the sarcoplasm contains many nuclei.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

196. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Hydrogen ions plays a very crucial role in the binding of oxygen with haemoglobin.
Reason R: Increase in the hydrogen ions in the blood will decrease the affinity of haemoglobin for oxygen.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) **A** is true but **R** is false.
(2) **A** is false but **R** is true.
(3) Both **A** and **R** are true and **R** is the correct explanation of **A**.
(4) Both **A** and **R** are true and **R** is NOT the correct explanation of **A**.

197. Match the columns and find out the correct combination:

	Column-I		Column-II
(a)	Nephridia	(i)	*Hydra*
(b)	Malpighian tubules	(ii)	Leech
(c)	Flame cells	(iii)	Shark
(d)	Kidneys	(iv)	Rotifers
		(v)	Cockroach

(1) (a)-(v) (b)-(ii) (c)-(iv) (d)-(iii)
(2) (a)-(ii) (b)-(v) (c)-(iv) (d)-(iii)
(3) (a)-(ii) (b)-(iv) (c)-(v) (d)-(i)
(4) (a)-(iv) (b)-(ii) (c)-(i) (d)-(v)

198. Given below are some statements **(A-D)**.
A. Cholecystokinin is a peptide hormone.
B. Secretin acts on the exocrine part of the pancreas.
C. Prolonged hyperglycemia leads to a complex disorder called diabetes insipidus.
D. Insulin enhances cellular glucose uptake and utilisation.
Choose the **correct** statements from the options given below:
(1) A and D only
(2) A and B only
(3) B, C and D only
(4) A, B and D only

199. The bulb-like structures present at the terminals of an axon are called the:
(1) synaptic knobs.
(2) cyton.
(3) synaptic vesicles.
(4) dendrites.

200. Which of the following statements is **incorrect**?
(1) Facial region in humans is formed from 14 skeletal elements.
(2) Malleus, incus and stapes collectively form pectoral girdle.
(3) Femur is the longest bone of our body.
(4) Acetabulum is a cavity formed at fusion point of ilium, ischium and pubis.